

Acetylcholine

1. Where in the CNS do we find somata of cholinergic neurons? Are there any in the PNS?
2. Draw the enzymatic synthesis and inactivation of acetylcholine.
Note: names of precursors, enzymes, co-factors, etc. are sufficient. It is NOT necessary to draw the chemical structures.
3. Describe / Draw a representation of the nicotinic ACh receptor found at the neuromuscular junction. How many subunits are required? Which kinds of subunits and how many of each? Where does the transmitter bind? What ion(s) permeate the activated receptor?
4. mACh receptors are G protein-coupled receptors (GPCR's), and there are 5 different types (M1 – M5). For each mAChR, identify the signal transduction mechanisms utilized by each subtype.

Receptor	M1	M2	M3	M4	M5
Effector Enzyme					
2 nd messenger system (including ↑ or ↓)					
Effect on ion channels					
Pharmacological effects					

5. Make a list of drugs and/or toxins that alter the cholinergic system

Mechanism	Example substances	Effect on cholinergic system
Synthesis		
Storage / Release		
Inactivation		
nAChR agonist		
nAChR antagonist		
mAChR agonist		
mAChR antagonist		